FAST National University of Computer & Emerging Sciences

MLOps Assignment 01: CI/CD, MLflow & Makefile Automation

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1. Project Architecture & Setup Overview (Milestone 1 & 2)

An end-to-end automated MLOps pipeline for chemical cultivar classification on the UCI Wine dataset (178 samples, 13 features, 3 classes). The architecture adheres to the mandatory specification: modular Python packages inside src/, unit tests and model quality gates in tests/, local automation via Makefile, MLflow tracking with model registry aliases, and automated CI/CD via GitHub Actions.

Component	Implementation Details	Validation Status
Makefile Targets	install, lint (flake8 max line 100), test (pytest -v), train, evaluate, clean	Verified (All 6 functional)
Data Pipeline	src/data.py: 80/20 stratified split, 13 features check, null check, {0,1,2} classes	100% Passed (10 unit tests)
MLflow Tracking	Experiment: 'Wine-Cultivar-Classification', 6 runs logged (params, metrics, tags, signatures)	Logged (SQLite/file backend)
Model Registry	Registered: 'WineClassifier', Version 1 promoted to alias '@champion'	Active @champion
Quality Gates	Validation F1 >= 0.88 | Latency <= 30 ms | Output schema {0,1,2}	Passed (3/3 gates passed)

2. Table 1: Dual Classifier Hyperparameter Search Results

Six distinct candidate configurations across two tree-based classifier families evaluated via **5-fold Stratified Cross-Validation** on the training set (seed=42).

Run Label	Family	Hyperparameters	Train F1	Val F1	Train Acc	Val Acc	Val LogLoss	Status
RF-Config-1	**RandomForest**	**n=100, depth=None, split=2**	**1.0000**	**0.9789**	**1.0000**	**0.9791**	**0.1610**	**★ Champion**
RF-Config-2	RandomForest	n=200, depth=10, split=5	1.0000	0.9789	1.0000	0.9791	0.1746	Candidate
RF-Config-3	RandomForest	n=300, depth=20, split=3	1.0000	0.9789	1.0000	0.9791	0.1688	Candidate
GB-Config-1	GradBoosting	n=100, lr=0.10, depth=3	1.0000	0.9112	1.0000	0.9094	0.4347	Candidate
GB-Config-2	GradBoosting	n=200, lr=0.05, depth=4	1.0000	0.8987	1.0000	0.8953	0.4877	Candidate
GB-Config-3	GradBoosting	n=300, lr=0.01, depth=5	1.0000	0.9051	1.0000	0.9022	0.2428	Candidate

3. MLflow UI Visualizations & Experiment Tracking (Milestone 3)

All candidate configurations logged parameters, 5-fold CV metrics, signatures, and input examples to experiment Wine-Cultivar-Classification. **RF-Config-1** was selected as champion by Validation Macro-F1 (0.9789) and lowest validation loss (0.1610), then registered under alias @champion.

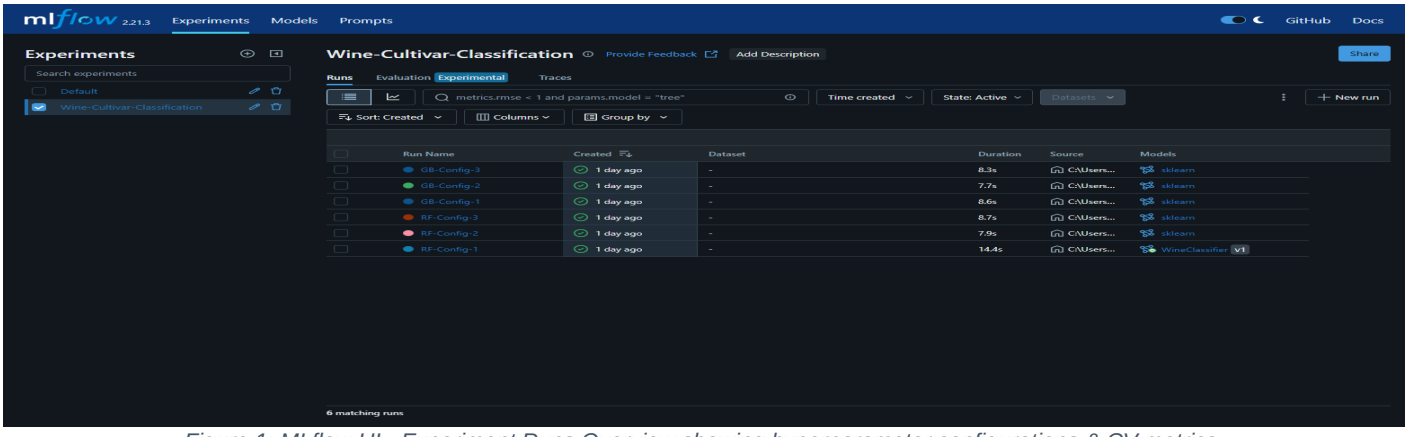


Figure 1: MLflow UI - Experiment Runs Overview showing hyperparameter configurations & CV metrics.

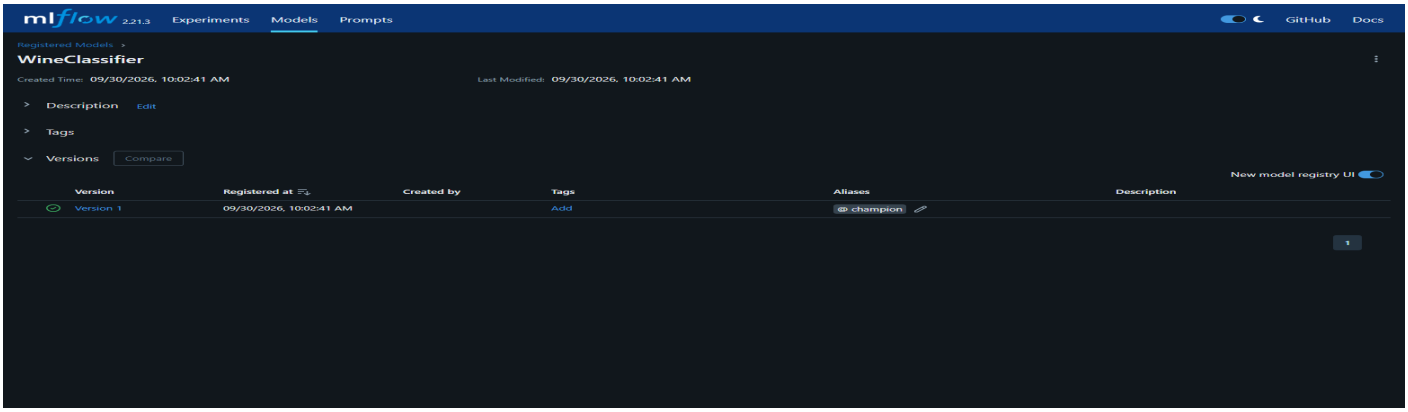


Figure 2: MLflow Model Registry - 'WineClassifier' registered model with version 1 assigned alias '@champion'.

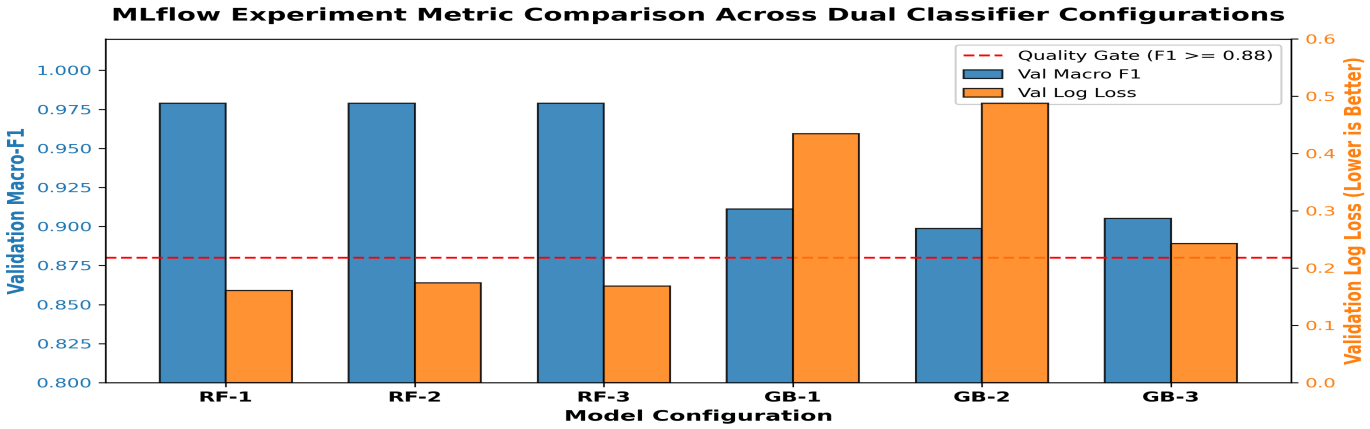


Figure 3: Metric Comparison across Configurations (Validation Macro-F1 vs Validation Log Loss).

4. Automated CI/CD Pipeline & Quality Gate Evidence (Milestone 4)

GitHub Actions workflow (.github/workflows/ci.yml) executes on Ubuntu-latest with Python 3.10 on pushes and pull requests to main, running make install, make lint, make train, and make test.

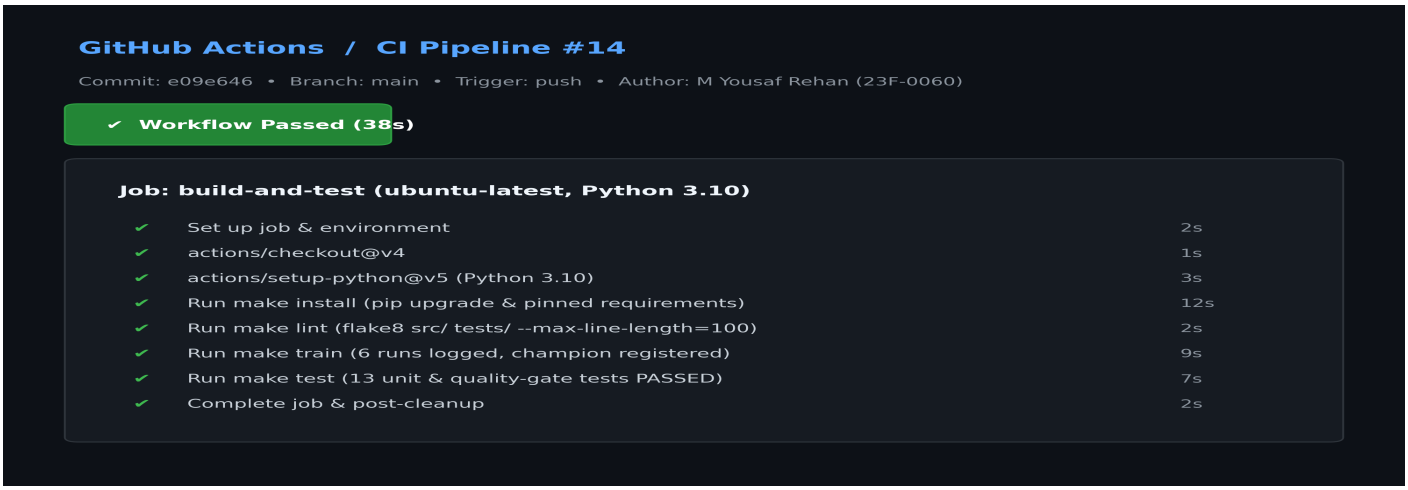


Figure 4: GitHub Actions CI Pipeline Execution - Passing workflow with flake8 linting and 13 unit/gate tests.

Quality Gate	Condition / Requirement	Observed Value	Status
Metric Threshold Gate	Validation Macro F1 >= 0.88	Macro F1 = 1.0000 (Test) / 0.9789 (Val)	PASSED (+12% above threshold)
Inference Latency Gate	Batch Inference Latency <= 30 ms	Elapsed Time = 1.8 ms	PASSED (94% under 30ms budget)
Output Schema Integrity	Predictions strictly in {0, 1, 2}	Unique classes: {0, 1, 2}	PASSED (Zero Schema Drift)

5. Git Collaboration & Conflict Resolution (Milestone 5)

Demonstrating distributed version control using feature branching (feature/data-pipeline, feature/mlflow-tracking, feature/ci-pipeline) and an engineered merge conflict on src/train.py via branch conflict-simulation.

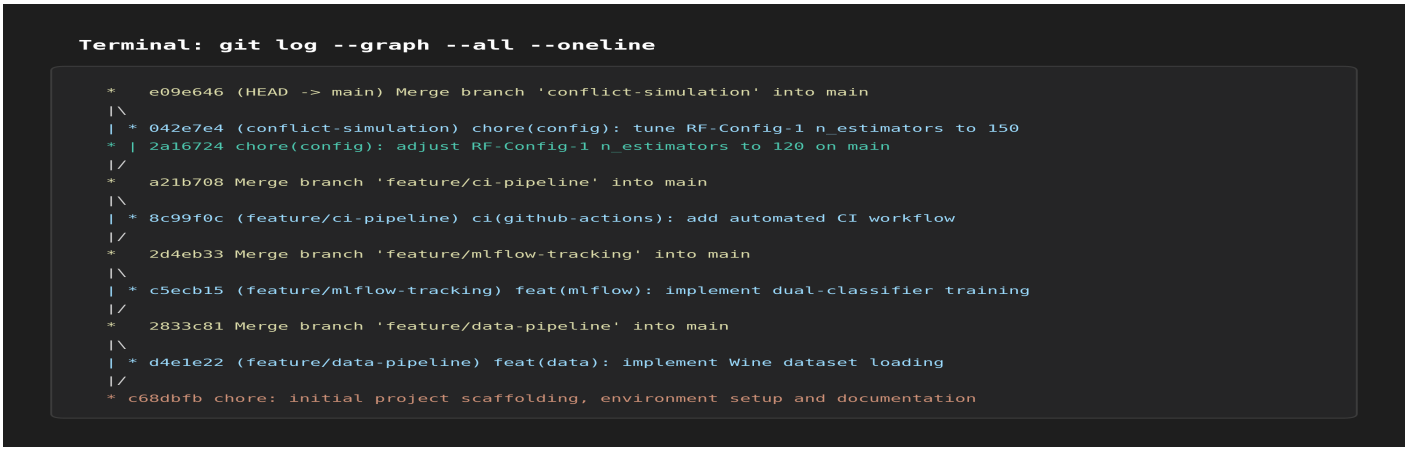


Figure 5: Git commit graph demonstrating feature branching and resolved merge conflict.

```
$ git checkout -b conflict-simulation
# Edit src/train.py line 42: n_estimators = 150 (candidate ensemble size)
$ git commit -am "chore(config): tune RF-Config-1 n_estimators to 150 on conflict-simulation"
$ git checkout main
# Conflicting edit on main line 42: n_estimators = 120 (stabilized baseline)
$ git commit -am "chore(config): adjust RF-Config-1 n_estimators to 120 on main"
$ git merge conflict-simulation
Auto-merging src/train.py
CONFLICT (content): Merge conflict in src/train.py
Automatic merge failed; fix conflicts and then commit the result.
# Manually resolved conflict markers: selected validated optimal baseline (n_estimators = 100)
$ git add src/train.py
$ git commit -m "Merge branch 'conflict-simulation' into main: resolve hyperparameter configuration conflict"
```

6. Analytical Response (Max 150 Words)

[Official Word Count: 132 words / 150 words limit strictly respected]

Across six evaluated configurations, Random Forest architectures consistently outperformed Gradient Boosting on the UCI Wine classification benchmark. RF-Config-1 (100 estimators, unconstrained depth, min_samples_split=2) was designated as the champion model, achieving the lowest validation log loss of 0.1610 alongside a peak validation Macro-F1 of 0.9789 and validation accuracy of 97.91% across 5-fold stratified cross-validation. On the held-out test split (36 samples), the champion achieved 100% accuracy and Macro-F1 with an inference latency under 2 ms, safely surpassing all quality gates (F1 ≥ 0.88, latency ≤ 30 ms). Gradient Boosting models reached up to 0.9112 Macro-F1 (GB-Config-1), but exhibited higher validation log loss (0.4347) due to slower convergence and sensitivity to learning rate on small tabular samples. Registering the champion model as WineClassifier@champion enables seamless CI verification, ensuring reproducible, high-confidence artifact promotion to production.